Conor Pedroche

business.pedroche@gmail.com linkedin.com/in/conor-pedroche/

Education

University of California, Berkeley Expected May 2028

B.A. in Computer Science | *Regents' and Chancellor's Scholar*

- **Relevant Coursework:** Intro to Programming with Python, Data Structures, Algorithm Analysis, Object-Oriented Programming, Functional Programming

Projects

Sourcecado | *Python*, *TypeScript*, *FastAPI*, *React*, *SQLite*, *pytest*, *Vitest*

- Developed CRM features for a local-first AI application for managing sponsor contacts, adding shared tasks across chats with descriptions, due dates, AI suggestions, and urgency labels.
- Implemented **Gmail synchronization** with persistent checkpoints, message deduplication, and contact matching, preserving conversation history and recovering synchronization progress after failures.
- Built **recovery safeguards** for interrupted AI workflows using saved checkpoints and execution records, identifying work that is safe to resume and requiring review before retrying external actions with unknown outcomes.

Doodle2Design | *Python*, *JavaScript*, *FastAPI*, *OpenAI API*, *pytest*

- Built a full-stack application that redesigns websites by translating user sketches, uploaded images, and visual preferences into redesigned web pages, winning **second place** in a hackathon hosted by Amazon and Google employees.
- Integrated OpenAI models for image analysis and website generation, using user inputs and comparing them to **30 design templates** to guide layout, style, and visual consistency.
- Created a dataset consisting of **50 images**, annotated with mood labels generated by models from OpenAI, Anthropic, and Google, retaining mood labels identified by all models.

Campus Event Finder | *Python*, *Flask*, *SQLite*, *HTML/CSS*, *pytest*

- Built a full-stack web application for students to search and filter campus events by date, category, and campus location.
- Designed a **Flask REST API** for event creation, retrieval, filtering, and deletion; used **SQLite** for persistent storage.
- Integrated **user authentication** and **input validation** to protect accounts and prevent invalid event submissions in the app.

Experience

UCSB Summer Research Academies Jun 2024 – Jul 2024

Research Intern *Santa Barbara, CA*

- Co-authored a research paper analyzing the connection between media outlet reputation and political bias among Reddit audiences and their interactions, presenting it at the Capstone Seminar.
- Collected and cleaned **thousands of Reddit discussion threads and comments** from hundreds of news outlets using the Reddit API, Selenium, and Python.
- Applied **VADER sentiment analysis**, correlation analysis, and **multiple linear regression** to study relationships between media reputation, article tone, political topics, audience responses, and audience interactions with other communities.

Desktop PC Builder Sep 2020 – Aug 2026

Builder / Technical Advisor *Santa Maria, CA*

- Designed, built, and optimized custom gaming PCs for **30+ customers**, selecting components based on budget, performance, and aesthetics, using PCPartPicker to research and compare CPUs, GPUs, PSUs, cooling systems, and other components.
- Diagnosed and resolved hardware and software failures, including POST errors, boot loops, overheating, memory instability, driver conflicts, system crashes, and BSODs through BIOS/UEFI configuration, component isolation, and stress testing.

Technical Skills

Languages: Python, C, SQL, TypeScript, JavaScript, HTML/CSS
Technologies/Frameworks: React, FastAPI, Flask, SQLite, Selenium, REST APIs
Developer Tools and Testing: Vitest, pytest, Git, GitHub

Honors & Awards

SkillsUSA Cybersecurity — State and Regional Gold Medalist